Worked Key: Unit 2 Practice Test

1. (2.1–2.2) Linear, exponential or neither?

Answer: (a) linear (differences all 3); (b) exponential (ratios all 2); (c) neither (differences 3, 5, 7, 9 are not constant and ratios are not constant). (d) $g(x) = 3 \cdot 2^x$.

Check **differences** first (constant means linear), then **ratios** (constant means exponential).

(a) $5-2=3,\ 8-5=3,\ 11-8=3,\ 14-11=3$. Constant differences, so **linear**: $f(x) = 3x + 2$.

(b) Differences are $3, 6, 12, 24$: not constant. Ratios: $\frac{6}{3} = \frac{12}{6} = \frac{24}{12} = \frac{48}{24} = 2$. Constant ratio, so **exponential**.

(c) Differences are $3, 5, 7, 9$ (not constant). Ratios are $4, 2.25, 1.78, 1.56$ (not constant). **Neither.** The *second* differences are constant (2), so it is quadratic: $h(x) = (x+1)^2$.

(d) $g(0) = 3$ is the starting value a, and the ratio is $b = 2$: $\boxed{g(x) = 3 \cdot 2^x}$.

2. (2.3) Exponential function features

Answer: (a) decay, 15% decrease; (b) ≈ 122.8; (c) $\to 0$ as $x \to \infty$, $\to \infty$ as $x \to -\infty$; (d) $y = 0$.

(a) The base is $b = 0.85 < 1$, so it is **decay**. $0.85 = 1 - 0.15$, so each step the value drops by **15%**.

(b) $f(3) = 200(0.85)^3 = 200(0.614125) \approx \boxed{122.8}$.

(c) Multiplying by 0.85 repeatedly shrinks toward 0, so $\lim_{x\to\infty} f(x) = 0$. Going left, we divide by 0.85 over and over, so the values grow without bound: $\lim_{x\to-\infty} f(x) = \infty$.

(d) The graph approaches 0 on the right but never reaches it: horizontal asymptote $\boxed{y = 0}$.

3. (2.4) Exponential model in context

Answer: (a) $P(t) = 50000(0.97)^t$; (b) $\approx 36{,}871$; (c) $t = \dfrac{\ln 0.5}{\ln 0.97} \approx 22.8$ years.

(a) Losing 3% means keeping 97%: growth factor $b = 1 - 0.03 = 0.97$. $\boxed{P(t) = 50000(0.97)^t}$.

(b) $P(10) = 50000(0.97)^{10} \approx \boxed{36{,}871}$ people.

(c) Half is 25,000. Solve $50000(0.97)^t = 25000$. Divide: $0.97^t = 0.5$. Take logs: $t \ln 0.97 = \ln 0.5$, so $t = \dfrac{\ln 0.5}{\ln 0.97} \approx \boxed{22.8 \text{ years}}$.

4. (2.5) Choosing a model

Answer: (a) Model B: its residuals are scattered with no pattern. Model A's residuals form a U-shape, so a line is the wrong kind of function. (b) Residual = actual − predicted, so the model overestimated that point by 3.

(a) A good model has residuals that scatter randomly around 0. Model A's residuals go $+, +, -, -, -, +, +$: a clear **U-shaped curve**, which tells us the data curve and a line cannot follow it. Model B's residuals bounce around with no pattern, so **Model B** is the better choice.

(b) residual = actual − predicted. A residual of -3 means the actual value was 3 *below* what the model predicted (the model overestimated).

5. (2.6) Composition

Answer: (a) 3; (b) $g(f(x)) = 4x^2 + 4x - 2$; (c) No: $f(g(x)) = 2x^2 - 5$ is different.

(a) Inside first: $g(2) = 4 - 3 = 1$. Then $f(1) = 3$. $\boxed{3}$.

(b) $g(f(x)) = (2x+1)^2 - 3 = 4x^2 + 4x + 1 - 3 = \boxed{4x^2 + 4x - 2}$.

(c) $f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 5$. That is not $4x^2 + 4x - 2$, so **no**: order matters. (Check with $x = 0$: $f(g(0)) = -5$ but $g(f(0)) = -2$.)

6. (2.7) Inverse functions

Answer: (a) $f^{-1}(x) = \dfrac{x+6}{3}$; (b) 2 and 5; (c) No: it fails the horizontal line test ($h(2) = h(-2)$).

(a) Write $y = 3x - 6$, swap: $x = 3y - 6$. Solve: $3y = x + 6$, so $\boxed{f^{-1}(x) = \dfrac{x+6}{3}}$.

(b) An inverse swaps inputs and outputs. $f(2) = 7$ means $f^{-1}(7) = \boxed{2}$. $f(5) = -1$ means $f^{-1}(-1) = \boxed{5}$.

(c) No. Two different inputs give the same output ($h(2) = h(-2) = 4$), so h is not one-to-one, and an inverse would have to send 4 to two different values. (A horizontal line at $y = 4$ crosses the graph twice.)

7. (2.8) Evaluate without a calculator

Answer: (a) 5; (b) -2; (c) -3; (d) 4; (e) -3.

Rewrite $\log_b(x) = y$ as the question **"what power of b gives x?"**

(a) $2^5 = 32$, so $\boxed{5}$. (b) $5^{-2} = \frac{1}{25}$, so $\boxed{-2}$. (c) $\left(\frac{1}{2}\right)^{-3} = 2^3 = 8$, so $\boxed{-3}$. (d) ln is base e, and $e^4 = e^4$, so $\boxed{4}$. (e) log is base 10, and $10^{-3} = 0.001$, so $\boxed{-3}$.

8. (2.10) A logarithmic function

Answer: (a) domain $x > 2$; vertical asymptote $x = 2$; (b) $g(5) = 2$; (c) $x = \dfrac{7}{3}$.

(a) The input of a log must be positive: $x - 2 > 0$, so the domain is $\boxed{x > 2}$. The graph blows down toward the line $\boxed{x = 2}$ (vertical asymptote).

(b) $g(5) = \log_3(3) + 1 = 1 + 1 = \boxed{2}$.

(c) Set $g(x) = 0$: $\log_3(x - 2) = -1$. Rewrite as a power: $x - 2 = 3^{-1} = \frac{1}{3}$, so $x = \boxed{\dfrac{7}{3}}$.

9. (2.11) Properties of logarithms

Answer: (a) $3 + 3\log_2 x - \log_2 y$; (b) $\log \dfrac{4x^3}{y^2}$; (c) False: the product rule says $\log(ab) = \log a + \log b$; counterexample $a = b = 1$: $\log 2 \neq 0$.

(a) $\log_2(8x^3) - \log_2 y = \log_2 8 + \log_2 x^3 - \log_2 y = \boxed{3 + 3\log_2 x - \log_2 y}$.

(b) Powers go back up: $\log x^3 - \log y^2 + \log 4$. Plus means multiply, minus means divide: $\boxed{\log \dfrac{4x^3}{y^2}}$.

(c) **False.** The rule is for *products*: $\log(ab) = \log a + \log b$. Counterexample: $a = b = 1$ gives $\log 2 \approx 0.30$ but $\log 1 + \log 1 = 0$.

10. (2.12) Solving equations

Answer: (a) $x = \frac{5}{2}$; (b) $x = \dfrac{\ln 20}{\ln 7} \approx 1.54$; (c) $x = 4$ (reject -2); (d) $x = e^2 - 1 \approx 6.39$.

(a) $81 = 3^4$, so $2x - 1 = 4$ and $\boxed{x = \frac{5}{2}}$.

(b) 20 is not a power of 7, so take a log of both sides: $x \ln 7 = \ln 20$, so $\boxed{x = \dfrac{\ln 20}{\ln 7} \approx 1.54}$.

(c) Condense: $\log_2[x(x-2)] = 3$, so $x(x-2) = 2^3 = 8$. Then $x^2 - 2x - 8 = 0$, which factors as $(x-4)(x+2) = 0$. Check the domain: both $x > 0$ and $x - 2 > 0$ are needed, so $x = -2$ is **extraneous**. $\boxed{x = 4}$.

(d) Rewrite as a power of e: $x + 1 = e^2$, so $\boxed{x = e^2 - 1 \approx 6.39}$.

11. (2.13–2.14) Half-life and semi-log

Answer: (a) $A(t) = 80\left(\frac{1}{2}\right)^{t/6}$; (b) ≈ 14.1 g; (c) $t = 24$ days; (d) a straight line with slope $-\frac{1}{6}$.

(a) Every 6 days the mass is multiplied by $\frac{1}{2}$: $\boxed{A(t) = 80\left(\frac{1}{2}\right)^{t/6}}$.

(b) $A(15) = 80\left(\frac{1}{2}\right)^{2.5} \approx \boxed{14.1 \text{ g}}$.

(c) $80\left(\frac{1}{2}\right)^{t/6} = 5$, so $\left(\frac{1}{2}\right)^{t/6} = \frac{1}{16} = \left(\frac{1}{2}\right)^4$. Then $\frac{t}{6} = 4$ and $\boxed{t = 24 \text{ days}}$. (That is four halvings: $80 \to 40 \to 20 \to 10 \to 5$.)

(d) $\log_2 A = \log_2 80 + \frac{t}{6}\log_2 \frac{1}{2} = \log_2 80 - \frac{t}{6}$. This is linear in t: a **straight line** with slope $\boxed{-\frac{1}{6}}$. A straight line on a semi-log plot is the signature of an exponential function.